

The Collected Mathematical Papers of Arthur Cayley

Volume V

Pages 177–201 (PDF pp. 201–225)

Arthur Cayley

Contents of this excerpt

Paper 339 (concluded) “On Skew Surfaces, otherwise Scrolls” (pp. 177–200)

Paper 340 (begins) “A Second Memoir on Skew Surfaces, otherwise Scrolls” (p. 201)

339. On Skew Surfaces, otherwise Scrolls (*continued*)

[Continuation from p. 176, in the middle of the investigation of the G formulae.]

25. $G(m^2, n^2)$.—I investigate the value by a process similar to that employed for $G(1, 1, m^2)$. Suppose that the curves m and n are plane curves having respectively h and k double points; then the line of intersection of the two planes meets the curve m in m points, and the curve n in n points; or, combining in every manner the m points two and two together, and the n points two and two together, the line in question is to be considered as $\frac{1}{2}[m]^2 \cdot \frac{1}{2}[n]^2$ coincident lines, each meeting the curve m twice and the curve n twice. There are besides the hk lines joining each double point of the curve m with each double point of the curve n . This gives in all $\frac{1}{4}[m]^2[n]^2 + hk$ lines; or, writing $h = \frac{1}{2}[m]^2 + M$, $k = \frac{1}{2}[n]^2 + N$, the number is

$$= \frac{1}{2}[m]^2[n]^2 + M \cdot \frac{1}{2}[n]^2 + N \cdot \frac{1}{2}[m]^2 + MN;$$

which is the value of $G(m^2, n^2)$ given by the investigation.

26. $G(m^2, n)$.—We have

$$G(m^2, n) = n G(1, m^2) = n S(m^2),$$

and it is in fact the same question to find $G(1, m^2)$ and to find $S(m^2)$. I assume for the present that the value [of $S(m^2)$, see post Art. 38] is $= \frac{1}{2}[m]^3 + M(m-2)$; and we then have

$$G(m^2, n) = n \left(\frac{1}{2}[m]^3 + M(m-2) \right).$$

27. Before going further, I observe that there are certain functional conditions which must be satisfied by the G formulae. Thus if the curve m be replaced by the system of the two curves

m, m' , instead of M we have $M + M'$. Let $G(m)$ denote any one of the functions $G(m, n, p, q)$, $G(m, n^2, p)$, $G(m, n^3)$; we must have

$$G(m + m') = G(m) + G(m').$$

Similarly, if $G(m^2)$ denote either of the functions $G(m^2, n, p)$, $G(m^2, n^2)$, we must have

$$G(m + m')^2 = G(m^2) + G(m, m') + G(m'^2);$$

and so if $G(m^3)$ stand for $G(m^3, n)$, then

$$G(m + m')^3 = G(m^3) + G(m^2, m') + G(m, m'^2) + G(m'^3);$$

and finally

$$G(m + m')^4 = G(m^4) + G(m^3, m') + G(m^2, m'^2) + G(m, m'^3) + G(m'^4).$$

28. The first three equations may be at once verified by means of the above given values of the G functions. But conversely, at least on the assumption that $G(m)$, $G(m^2)$, &c., in so far as they respectively depend on the curve m , are functions of m and M only, we may, by the solution of the functional equations, obtain the values of the G functions. It is to be observed that the first equation is of the form

$$\phi(m + m') = \phi(m) + \phi(m'),$$

the general solution whereof is

$$\phi m = \alpha m + \beta M;$$

the second equation, supposing that $G(m, m')$ is known—the third equation, supposing that $G(m^2, m')$ and $G(m, m'^2)$ are known—and the fourth equation, supposing that $G(m^3, m')$, $G(m^2, m'^2)$, $G(m, m'^3)$ are known, are respectively of the form

$$\phi(m + m') = \phi m + \phi m' + \text{Funct.}(m, m');$$

and hence if a particular solution be given, the general solution is

$$\phi(m) = \text{Particular Solution} + \alpha m + \beta M.$$

The values of the constants must in each case be determined by special considerations.

29. The value of $G(m, n, p, q)$ was obtained strictly; that of $G(m^2, n, p)$ was reduced to depend on $G(1, 1, m^2)$, and that of $G(m^2, n)$ on $G(1, m^2)$. I apply therefore the functional equations to the confirmation of the values of $G(1, 1, m^2)$, $G(m^2, n^2)$, and $G(1, m^2)$, and to the determination of the value of $G(m^4)$.

30. First, if $G(m^2)$ denote $G(1, 1, m^2)$, then $G(m, m')$ denotes $G(1, 1, m, m')$, which is $= 2mm'$; hence

$$G(m + m')^2 - G(m^2) - G(m'^2) = 2mm',$$

which is satisfied by $G(m^2) = [m]^2$. This gives

$$G(1, 1, m^2) = [m]^2 + \alpha m + \beta M;$$

but if the curve m be a system of m lines ($m = m, M = 0$), then $G(1, 1, m^2) = [m]^2$; and again, if the curve m be a conic ($m = 2, M = -1$), then $G(1, 1, m^2) = 1$. This gives $\alpha = 0$, $\beta = 1$, and therefore

$$G(1, 1, m^2) = [m]^2 + M.$$

31. Next, if $G(m^2)$ denote $G(m^2, n^2)$, then $G(m, m')$ denotes $G(m, m', n^2)$, which is $= mm'([n]^2 + N)$. The functional equation is

$$G(m + m')^2 - G(m^2) - G(m'^2) = mm'([n]^2 + N),$$

which is satisfied by $G(m^2) = \frac{1}{2}[m]^2([n]^2 + N)$. Hence we have

$$G(m^2, n^2) = \frac{1}{2}[m]^2([n]^2 + N) + \alpha m + \beta M,$$

where α, β are functions of n, N ; and observing that $G(m^2, n^2)$ must be symmetrical in regard to the curves m and n , it is easy to see that we may write

$$G(m^2, n^2) = \frac{1}{2}[m]^2[n]^2 + M \cdot \frac{1}{2}[n]^2 + N \cdot \frac{1}{2}[m]^2 + \alpha mn + \beta(mN + nM) + \gamma MN,$$

where α, β, γ are absolute constants. To determine them, if the curve m be a pair of lines ($m = 2, M = 0$), then

$$G(m^2, n^2) = G(1, 1, n^2) = [n]^2 + N;$$

and if each of the curves m, n be a conic ($m = 2, M = -1, n = 2, N = -1$), then

$$G(m^2, n^2) = 1.$$

These cases give $\alpha = \beta = 0, \gamma = 1$, and therefore

$$G(m^2, n^2) = \frac{1}{2}[m]^2[n]^2 + M \cdot \frac{1}{2}[n]^2 + N \cdot \frac{1}{2}[m]^2 + MN.$$

32. Again, $G(m^2)$ standing for $G(1, m^3)$, then $G(m^2, m')$ and $G(m, m'^2)$ will stand for $G(1, m^2, m')$ and $G(1, m, m'^2)$, the values whereof are $m'([m]^2 + M)$ and $m([m']^2 + M')$ respectively. We have thus

$$G(m + m')^3 - G(m^3) - G(m'^3) = m'([m]^2 + M) + m([m']^2 + M'),$$

a solution of which is $G(m^3) = \frac{1}{3}[m]^3 + mM$. Hence we have

$$G(1, m^3) = \frac{1}{3}[m]^3 + mM + \alpha m + \beta M.$$

Suppose first that the curve m is a system of lines ($m = m, M = 0$), then $G(1, m^3) = \frac{1}{3}[m]^3$; and next that the curve m is a cubic in space or skew cubic ($m = 3, M = -2$), then $G(1, m^3) = 0$, since a line can meet the curve in two points only. We thus find $\alpha = 0, \beta = -2$, and thence

$$G(1, m^3) = \frac{1}{3}[m]^3 + M(m - 2).$$

33. Hence, substituting for $G(m^3, m')$, $G(m^2, m'^2)$, $G(m, m'^3)$ their values $m'(\frac{1}{3}[m]^3 + M(m - 2))$, $\frac{1}{2}[m]^2[m']^2 + M \cdot \frac{1}{2}[m']^2 + M' \cdot \frac{1}{2}[m]^2 + MM'$, $m(\frac{1}{3}[m']^3 + M'(m' - 2))$ respectively, we find

$$\begin{aligned} G(m + m')^4 - G(m^4) - G(m'^4) &= m'(\frac{1}{3}[m]^3 + M(m - 2)) \\ &\quad + \frac{1}{2}[m]^2[m']^2 + M \cdot \frac{1}{2}[m']^2 + M' \cdot \frac{1}{2}[m]^2 + MM' \\ &\quad + m(\frac{1}{3}[m']^3 + M'(m' - 2)), \end{aligned}$$

and thence, obtaining first a particular solution, the general solution is

$$G(m^4) = \frac{1}{12}[m]^4 + M(\frac{1}{2}[m]^2 - 2m) + M^2 + \alpha m + \beta M.$$

34. To determine the constants, suppose first that the curve m is a system of lines ($m = m, M = 0$), we must have $G(m^4) = \frac{1}{12}[m]^4 + m$, and thence $\alpha = 0$. Next, if the curve m be a conic ($m = 2, M = -1$), we must have $G(m^4) = 0$; and this gives $\beta = \frac{1}{2}$, and consequently

$$G(m^4) = \frac{1}{12}[m]^4 + m + M(\frac{1}{2}[m]^2 - 2m + \frac{1}{2}) + M^2.$$

The NG formulae, Article 35.

35. The *NG* formulae are now at once obtained, viz. we have

$$\begin{aligned} NG(m^2, n, p) &= G(m^2, n, p) + G(m, n^2, p) + G(m, n, p^2), \\ NG(m^3, n) &= 3G(m^3, n) + G(m^2, n^2), \\ NG(m^4) &= 6G(m^4), \end{aligned}$$

which give the values in the Table.

The S formulae, particular cases, Articles 36 to 40.

36. The *S* formulae have in fact been obtained in the investigation of the *G* formulae: we have

$$\begin{aligned} S(m, n, p) &= 2mnp, \\ S(m^2, n) &= n([m]^2 + M), \\ S(m^3) &= \frac{1}{3}[m]^3 + M(m - 2). \end{aligned}$$

37. In confirmation of the formula $S(1, m^2) = [m]^2 + M$, it is to be remarked that if we take through the line 1 an arbitrary plane, this meets the curve m in m points, and joining these two and two together we have $\frac{1}{2}[m]^2$ lines, each of them meeting the curve m twice and also meeting the line 1; that is, the lines in question are generating lines of the scroll $S(1, m^2)$. The line 1 is, as already mentioned, an $(h = (\frac{1}{2}[m]^2 + M))$ tuple line on the scroll; the section by the arbitrary plane is therefore the line 1 taken $(\frac{1}{2}[m]^2 + M)$ times, together with the before-mentioned $\frac{1}{2}[m]^2$ lines; that is, the order of the surface is $[m]^2 + M$, as it should be. This is in fact the mode in which the order of the scroll $S(1, m^2)$ was originally obtained by Dr Salmon.

38. As regards the formula $S(m^3) = \frac{1}{3}[m]^3 + M(m - 2)$, suppose that the curve m is a (p, q) curve on the hyperboloid; we have as before $m = p + q$, $M = -pq$, and the formula becomes

$$S(m^3) = \frac{1}{3}[p + q]^3 - pq(p + q - 2),$$

which is

$$= \frac{1}{3}[p]^3 + \frac{1}{3}[q]^3$$

as already remarked, the surface is in this case the hyperboloid taken $\frac{1}{3}[p]^3 + \frac{1}{3}[q]^3$ times.

39. It is to be noticed also that if the curve m be a system of lines ($m = m$, $M = 0$), then the formula gives

$$S(m^3) = \frac{1}{3}[m]^3,$$

which is right, since in this case the scroll is made up of the $\frac{1}{3}[m]^3$ hyperboloids generated each of them by a line which meets three out of the m lines.

In the case of a curve m , which is such that the coordinates of any point of the curve are proportional to rational and integral functions of the order m of an arbitrary parameter θ , or say the case of a unicursal curve of the order m , we have

$$(h = \frac{1}{2}[m - 1]^2 \text{ and } \therefore) M = -(m - 1),$$

and the formula gives

$$S(m^3) = \frac{1}{3}[m]^3 - (m - 1)(m - 2),$$

for a direct investigation of which see post, Annex No. 1.

40. In the case of a curve m , which is the complete intersection of two surfaces of the orders p and q respectively, or say a complete $(p \times q)$ intersection, we have

$$m = pq, \quad (h = \frac{1}{2}pq(p-1)(q-1) \text{ and } \therefore) M = -\frac{1}{2}pq(p+q-2);$$

and we find

$$\begin{aligned} S(m^3) &= \frac{1}{6}pq(pq-2)(2pq-3p-3q+4), \\ &= \frac{1}{6}\beta(\beta-2)(2\beta-3\alpha+4) \end{aligned}$$

if $\alpha = pq$, $\beta = p+q$. The mode of obtaining this result by a direct investigation was pointed out to me by Dr Salmon; see post, Annex No. 2.

Particular cases of the formula for $G(m^4)$, Articles 41 & 42.

41. In the case of a (p, q) curve on the hyperboloid, putting as before $m = p+q$, $M = -pq$, we find

$$G(m^4) = \frac{1}{12}[p+q]^4 + p+q - pq(\frac{1}{2}[p+q]^2 - 2(p+q) + \frac{1}{2}) + p^2q^2,$$

which is

$$= \frac{1}{12}([p]^4 + [q]^4) - 2p[q-1]^2 - 2q[p-1]^2,$$

vanishing if p, q are neither of them greater than 3: this is as it should be, since there is then no line which meets the curve four times. The curves for which the condition is satisfied are (1, 1) the conic, (1, 2) the cubic, (2, 2) the quadriquadric, (1, 3) the excubo-quartic, (2, 3) the excubo-quintic (viz. the quintic curve, which is the partial intersection of a quadric surface and a cubic surface having a line in common), and (3, 3) the quadri-cubic, or complete intersection of a quadric surface and a cubic surface. If either p or q exceeds 3, we have the case of a curve through every point whereof there can be drawn a line or lines through four or more points, and the formula is inapplicable.

42. In the case of a complete $(p \times q)$ intersection, we have as before $m = pq$, $M = -\frac{1}{2}pq(p+q-2)$, and the formula for $G(m^4)$ becomes

$$G(m^4) = \frac{1}{12}\beta \left\{ 2\beta^3 - 6\alpha\beta^2 + \beta(3\alpha^2 + 18\alpha - 26) - 66\alpha + 144 \right\},$$

a formula the direct verification whereof is due to Dr Salmon; see post, Annex No. 3.

The formulae for $NR(1, m, n)$ and $NR(1, m^2)$, Articles 43 to 46.

43. $NR(1, m, n)$.—Through the line 1 take any plane meeting the curve m in m points and the curve n in n points; then if m_1, m_2 be any two of the m points, and n_1, n_2 any two of the n points, the lines m_1n_1 and m_2n_2 are generating lines of the scroll $S(1, m, n)$, and these lines intersect in a point which belongs to the Nodal Residue NR ; and in like manner the lines m_1n_2 and m_2n_1 are generating lines of the scroll, and they intersect on a point of NR ; we have thus

$$(2 \cdot \frac{1}{2}[m]^2 \cdot \frac{1}{2}[n]^2 =) \frac{1}{2}[m]^2[n]^2$$

points on NR , that is, the arbitrary plane through the line 1 cuts NR in $\frac{1}{2}[m]^2[n]^2$ points. But the plane also cuts NR in certain points lying on the line 1, and if the number of these be (a) , then

$$NR(1, m, n) = \frac{1}{2}[m]^2[n]^2 + a.$$

44. The points (a) are included among the cuspidal points on the line 1. Taking for a moment $x = 0, y = 0$ for the equations of the line 1 (which, as we have seen, is a mn -tuple line

on the scroll), the equation of the scroll is of the form $(A, \dots | x, y)^{mn} = 0$, where $A, \dots$ are functions of the coordinates of the degree mn . The entire number of cuspidal points on the line 1 is thus $= 2[mn]^2$; but these include different kinds of cuspidal points, viz. we have

$$2[mn]^2 = 2a + 2\alpha + 2\alpha' + R,$$

if (a) be the number of points in which the line 1 meets NR ,

$$\begin{array}{llll} \alpha & \text{'' '' '' ''} & S(m^2, n), \\ \alpha' & \text{'' '' '' ''} & S(m, n^2), \\ R & \text{'' '' '' ''} & \text{Torse}(m, n), \end{array}$$

where by $\text{Torse}(m, n)$ I denote the developable surface or “Torse” generated by a line which meets each of the curves m and n . The order of the Torse in question is

$$R = (n([m]^2 - 2h) + m([n]^2 - 2k) =) 2(nM + mN),$$

see post, Annex No. 4. And then observing that we have

$$2a + 2\alpha + 2\alpha' + R = 2m[m]^2 + 2m[n]^2 \quad (\text{as above}),$$

these values give

$$\begin{aligned} \alpha &= S(m^2, n) = n([m]^2 + M), \\ \alpha' &= S(m, n^2) = m([n]^2 + N), \end{aligned}$$

and we have

$$\begin{aligned} a &= \frac{1}{2}(2[mn]^2 - 2\alpha - 2\alpha' - R), \\ &= [m]^2[n]^2 - n[m]^2 - m[n]^2, \end{aligned}$$

and thence

$$NR(1, m, n) = \frac{3}{2}[m]^2[n]^2.$$

45. $NR(1, m^2)$.—Through the line 1 take any arbitrary plane meeting the curve m in m points; if m_1, m_2, m_3, m_4 be any four of these, then the lines m_1m_2 and m_3m_4 are generating lines of the scroll $S(1, m^2)$, and their intersection is a point of the nodal residue NR ; but in like manner the lines m_1m_3 and m_2m_4 are generating lines of the scroll, and their intersection is a point of NR ; and so the lines m_1m_4 and m_2m_3 are generating lines of the scroll, and their intersection is a point of NR . We have thus $(3 \times \frac{1}{4}[m]^4 =) \frac{3}{4}[m]^4$ points of NR on the arbitrary plane through the line 1. But there are besides the points of NR which lie on the line 1; and if the number of these be (a) , then

$$NR(1, m^2) = \frac{3}{4}[m]^4 + a.$$

46. The points (a) are included among the cuspidal points of the scroll lying on the line 1. Supposing for a moment that $x = 0, y = 0$ are the equations of the line 1, then this line being a $(\frac{1}{2}[m]^2 + M)$ -tuple line on the scroll, the equation of the scroll is of the form $(A, \dots | x, y)^{\frac{1}{2}[m]^2 + M} = 0$, where $A, \dots$ are functions of the coordinates of the degree $\frac{1}{2}[m]^2$: the number of cuspidal points on the line 1 is thus

$$(2 \cdot \frac{1}{2}[m]^2 (\frac{1}{2}[m]^2 - 1 + M) =) [m]^2 (\frac{1}{2}[m]^2 - 1 + M).$$

But these include cuspidal points of several kinds, viz. we have

$$[m]^2 (\frac{1}{2}[m]^2 - 1 + M) = 2a + 3\beta + R',$$

$$\begin{array}{ll}
\text{if } (a) & \text{be the number of points in which the line 1 meets } NR, \\
\beta & \text{„ „ „ „ } S(m^3), \\
R' & \text{„ „ „ „ } \text{Torse}(m^2),
\end{array}$$

where $\text{Torse}(m^2)$ denotes the developable surface or Torse generated by a line which meets the curve m twice. The order of the Torse in question is

$$R' = -2(m-3)M$$

(see post, Annex No. 5); and then since $\beta = S(m^3) = \frac{1}{3}[m]^3 + M(m-2)$, we find

$$2a = [m]^2 \left(\frac{1}{2}[m]^2 - 1 + M \right) - 3 \left(\frac{1}{3}[m]^3 + M(m-2) \right) + 2M(m-3),$$

and thence

$$\begin{aligned}
a &= \frac{1}{4}[m]^4 + \frac{1}{2}[m]^3 + M([m]^2 - m), \\
NR(1, m^2) &= \frac{3}{4}[m]^4 + \frac{1}{2}[m]^3 + M\left(\frac{1}{2}[m]^2 - \frac{1}{2}m\right).
\end{aligned}$$

But I have not succeeded in finding by a like direct investigation the values of

$$NR(m, n, p), \quad NR(m^2, n), \quad NR(m^3).$$

Formulae for $NT(1, m, n)$, $NT(1, m^2)$, Articles 47 and 48.

47. We have

$$\begin{aligned}
NT(1, m, n) &= NG(1, m, n) &&= mn(m+n-2) + mN + nM \\
&+ ND(1, m, n) &&+ \frac{1}{2}mn(mn+m+n-3) \\
&+ NR(1, m, n) &&+ \frac{3}{2}[m]^2[n]^2,
\end{aligned}$$

which is

$$\begin{aligned}
&= 2[mn]^2 + mN + nM, \\
&= \frac{1}{2}S^2 - S + mN + nM,
\end{aligned}$$

where $S = S(1, m, n) = 2mn$.

48. And moreover

$$\begin{aligned}
NT(1, m^2) &= ND(1, m^2) &&= \frac{1}{4}[m]^4 + [m]^3 + M\left(\frac{1}{2}[m]^2 - \frac{1}{2}\right) + M^2 \cdot \frac{1}{2} \\
&+ NG(1, m^2) &&+ [m]^3 + M(3m-6) \\
&+ NR(1, m^2) &&+ \frac{3}{4}[m]^4 + M\left(\frac{1}{2}[m]^2 - 2m + 3\right),
\end{aligned}$$

which is

$$\begin{aligned}
&= \frac{1}{2}[m]^4 + 2[m]^3 + M([m]^2 + m - \frac{5}{2}) + M^2 \cdot \frac{1}{2}, \\
&= \frac{1}{2}S^2 - S + M\left(m - \frac{3}{2}\right),
\end{aligned}$$

if $S = S(1, m^2) = [m]^2 + M$.

The NT and NR formulae, Articles 49 to 58.

49. I proceed to find $NT(m, n, p)$, &c. by a functional investigation, such as was employed for finding $G(1, 1, m^2)$, &c. Writing $S(m)$ to denote either of the scrolls $S(m, n, p)$, $S(m, n^2)$, and supposing that in place of the curve m we have the aggregate of the two curves m, m' ; then the scroll $S(m + m')$ breaks up into the two scrolls $S(m)$, $S(m')$, and the intersection of these is part of the nodal total $NT(m + m')$; that is, we have

$$NT(m + m') = NT(m) + NT(m') + S(m) \cdot S(m');$$

and in like manner, if $S(m^2)$ stands for $S(m^2, n)$, then

$$NT(m + m')^2 = NT(m^2) + NT(m, m') + NT(m'^2) + C_1(S(m^2), S(m, m'), S(m'^2)),$$

where C_1 denotes the sum of the combinations two and two together; and so also

$$\begin{aligned} NT(m + m')^3 = & NT(m^3) + NT(m^2, m') + NT(m, m'^2) + NT(m'^3) \\ & + C_1(S(m^3), S(m^2, m'), S(m, m'^2), S(m'^3)). \end{aligned}$$

50. Instead of assuming

$$NT = \frac{1}{2}S^2 + \phi,$$

it is the same thing, and it is rather more convenient, to assume

$$NT = \frac{1}{2}S^2 - S + \phi,$$

viz. $NT(m) = \frac{1}{2}(S(m))^2 - S(m) + \phi(m)$, &c. Then observing that

$$S(m + m') = S(m) + S(m'), \text{ \&c.},$$

the foregoing equations for NT give

$$\begin{aligned} \phi(m + m') &= \phi(m) + \phi(m'), \\ \phi(m + m')^2 &= \phi(m^2) + \phi(m, m') + \phi(m'^2), \\ \phi(m + m')^3 &= \phi(m^3) + \phi(m^2, m') + \phi(m, m'^2) + \phi(m'^3); \end{aligned}$$

and if in the second equation $\phi(m, m')$ and in the third equation $\phi(m^2, m')$ and $\phi(m, m'^2)$ are regarded as known, these are all of them of the form

$$f(m + m') - f(m) - f(m') = \text{Funct.}(m, m');$$

so that, a particular solution being obtained, the general solution is $f(m) = \text{Particular Solution} + \alpha m + \beta M$, at least on the assumption that $f(m)$, in so far as it depends on the curve m , is a function of m and M only.

51. First, if $\phi(m)$ stands for $\phi(m, n, p)$, we obtain $\phi(m, n, p) = \alpha m + \beta M$, or observing that $\phi(m, n, p)$ must be symmetrical in regard to the curves m, n , and p , we may write

$$\phi(m, n, p) = \alpha mnp + \beta (Mnp + Nmp + Pmn) + \gamma (mNP + nMP + pMN) + \delta MNP,$$

and then

$$\begin{aligned} NT(m, n, p) &= \frac{1}{2}S^2 - S + \phi(m, n, p), \\ &= 2mnp(mnp - 1) + \phi(m, n, p). \end{aligned}$$

But for $p = 1$ this should reduce itself to the known value of $NT(1, m, n)$; this gives $\alpha = 0$, $\beta = 1$, $\gamma = 0$; we in fact have, as will be shown, post, Art. 55, $\delta = 0$; and hence

$$NT(m, n, p) = \frac{1}{2}S^2 - S + (Mnp + Nmp + Pmn),$$

$$= 2 [mnp]^2 + (Mnp + Nmp + Pmn).$$

52. Next, if $\phi(m^2)$ stand for $\phi(m^2, n)$, then $\phi(m, m')$ stands for $\phi(m, m', n)$, which is $= Nmm' + n(mM' + m'M)$, and the equation is

$$\phi(m + m')^2 - \phi(m^2) - \phi(m'^2) = Nmm' + n(mM' + m'M).$$

A particular solution is $\phi(m^2) = \frac{1}{2}[m]^2N + nmM$, and we have therefore

$$\phi(m^2, n) = \frac{1}{2}[m]^2N + nmM + \alpha m + \beta M;$$

or observing that $\phi(m^2, n)$ considered as a function of n , satisfies the equation

$$\phi(n + n') = \phi(n) + \phi(n'),$$

and is therefore a linear function of n and N , we may write

$$\phi(m^2, n) = \frac{1}{2}[m]^2N + nmM + \alpha nm + \beta nM + \gamma mN + \delta MN;$$

we then have

$$NT(m^2, n) = \frac{1}{2}S^2 - S + \phi(m^2, n),$$

where $S = S(m^2, n) = n([m]^2 + M)$.

And then putting $n = 1$, and comparing with the known value of $NT(1, m^2)$, we find $\alpha = 0$, $\beta = -\frac{3}{2}$. It will be shown, post, Art. 55, that $\gamma = 0$, $\delta = 0$; and we have therefore

$$\phi(m^2, n) = nM(m - \frac{3}{2}) + N(\frac{1}{2}[m]^2 + M),$$

and thence

$$\begin{aligned} NT(m^2, n) &= \frac{1}{2}S^2 - S + \phi(m^2, n), \\ &= n^2 \left(\frac{1}{2}[m]^4 + 2[m]^3 + M([m]^2 + m - \frac{3}{2}) + M^2 \cdot \frac{1}{2} \right) \\ &\quad + N^2 \left(\frac{1}{2}[m]^4 + 2[m]^3 + [m]^2 + M[m]^2 + M^2 \cdot \frac{1}{2} \right). \end{aligned}$$

53. Next for $\phi(m^4)$, substituting for $\phi(m^2, m')$ and $\phi(m, m'^2)$ their values, we have

$$\begin{aligned} \phi(m + m')^4 - \phi(m^4) - \phi(m'^4) &= m'M(m - \frac{3}{2}) + M'(\frac{1}{2}[m]^2 + M) \\ &\quad + mM'(m' - \frac{3}{2}) + M(\frac{1}{2}[m']^2 + M'), \end{aligned}$$

which is satisfied by

$$\phi(m^4) = M(\frac{1}{2}[m]^3 - \frac{3}{2}m) + M^2,$$

and the general value then is

$$\phi(m^4) = M(\frac{1}{2}[m]^3 - \frac{3}{2}m) + M^2 + \alpha m + \beta M,$$

and we have

$$NT(m^4) = \frac{1}{2}S^2 - S + \phi(m^4),$$

where

$$S = S(m^4) = \frac{1}{3}[m]^3 + M(m - 2).$$

54. Taking for the curve m the (p, q) curve on the hyperboloid ($m = p + q$, $M = -pq$), $S(m^4)$ becomes the hyperboloid taken k times, if $k = \frac{1}{3}[p]^3 + \frac{1}{3}[q]^3$; that is, $S(m^4) = 2k$, and

$NT(m^4) = 4 \cdot \frac{1}{2}[k]^2 + \phi(m^4)$; $\phi(m^4)$ must vanish if p and q are each not greater than 3; this implies $\alpha = 3$, $\beta = 11$, for with these values the formula gives

$$\phi(m^4) = -\frac{1}{6}(q[p-1]^3 + p[q-1]^3).$$

55. I assume the correctness of the value

$$\phi(m^4) = 3m + M(\frac{1}{2}[m]^3 - 4m + 11) + M^2$$

so obtained, as being in fact verified by means of the six several curves (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3); and I remark that if the foregoing value of $\phi(m, n, p)$ had been increased by δMNP , then it would have been necessary to increase the value of $\phi(m^2, n)$ by δM^2N , and that of $\phi(m^4)$ by δM^3 ; and moreover that if the foregoing value of $\phi(m^2, n)$ had been increased by $\gamma mN + \delta MN$, then it would have been necessary to increase the value of $\phi(m^4)$ by $\gamma mM + \delta M^2$; this is easily seen by writing

$$\begin{aligned}\phi(m^4) &= \gamma mM + \delta M^2 + \alpha M^3, \\ \phi(m^3, m') &= \gamma mM' + \delta MM' + 3\alpha M^2M', \\ \phi(m, m'^3) &= \gamma m'M + \delta MM' + 3\alpha MM'^2, \\ \phi(m'^4) &= \gamma m'M' + \delta M'^2 + \alpha M'^3,\end{aligned}$$

the sum of which is

$$= \gamma(m+m')(M+M') + \delta(M+M')^2 + \alpha(M+M')^3,$$

the corresponding term of $\phi(m^4)$; hence the value of $\phi(m^4)$ being correct without the foregoing addition, we must have $\gamma = 0$, $\delta = 0$, $\alpha = 0$; which confirms the foregoing values of $\phi(m, n, p)$, $\phi(m^2, n)$.

56. The equation

$$NT(m^4) = \frac{1}{2}S^2 - S + \phi(m^4)$$

gives

$$\begin{aligned}NT(m^4) &= \frac{1}{2}S^2 - S + 3m + M(\frac{1}{2}[m]^3 - 4m + 11) + M^2, \\ &= \frac{1}{18}[m]^6 + \frac{1}{3}[m]^3 + M^2 + 3m + \dots\end{aligned}$$

57. We have

$$\begin{aligned}NR(m^2, n) &= NT(m^2, n) - ND(m^2, n) - NG(m^2, n), \\ &= n(\frac{1}{4}[m]^4 + \frac{1}{2}[m]^3 + M(\frac{1}{2}[m]^2 - 2m + 3)) \\ &\quad + [n]^2(\frac{1}{4}[m]^4 + \frac{1}{2}[m]^3 + [m]^2 + M([m]^2 - \frac{1}{2}) + M^2 \cdot \frac{1}{2}).\end{aligned}$$

58. And moreover

$$\begin{aligned}NR(m^4) &= NT(m^4) - ND(m^4) - NG(m^4), \\ &= \frac{1}{72}[m]^6 + \frac{1}{6}[m]^4 - \frac{1}{6}[m]^3 + 3m \\ &\quad + M(\frac{1}{4}[m]^4 - \frac{1}{2}[m]^3 - \frac{1}{2}[m]^2 + 9m - 20) \\ &\quad + M^2(\frac{1}{2}[m]^2 - 2m);\end{aligned}$$

and the investigation of the series of results given in the Table is thus concluded.

Intersections of a generating line with the Nodal Total, Articles 59 to 63.

59. We may for the scrolls $S(1, m, n)$ and $S(1, m^2)$ verify the theorem that each generating line meets the Nodal Total in a number of points $= S - 2$.

In fact for the scroll $S(1, m, n)$, the directrix curves are respectively multiple curves of the orders mn, n, m , and a generating line meets each of these in a single point, counting for the three curves respectively as $mn - 1, n - 1$, and $m - 1$ points respectively. Moreover the construction (*ante*, Art. 43) for the Nodal Residue $NR(1, m, n)$ shows that a generating line meets this curve in $(m - 1)(n - 1)$ points; and since the curve is merely a double curve, these count each as a single point; and the generating line does not meet the Nodal Generator $NG(1, m, n)$. The number of intersections therefore is

$$mn - 1 + (m - 1) + (n - 1) + (m - 1)(n - 1),$$

which is

$$= 2mn - 2, \quad = S - 2.$$

60. Similarly for the scroll $S(1, m^2)$; the directrix curves are multiple curves, viz. the line 1 is a $(\frac{1}{2}[m]^2 + M)$ -tuple curve, and the curve m a $(m - 1)$ -tuple curve; the generating line meets the former in a single point, counting as $\frac{1}{2}[m]^2 + M - 1$ points, and the latter in two points, each counting as $(m - 2)$ points. The construction (*ante*, Art. 45) for the Nodal Residue $NR(1, m^2)$ shows that the generating line meets this curve in $\frac{1}{2}[m - 2]^2$ points; and since the curve is merely a double curve, these count each as a single point. Finally, the generating line does not meet the Nodal Generator $NG(1, m^2)$. The number of intersections thus is

$$\frac{1}{2}[m]^2 - 1 + M + 2(m - 2) + \frac{1}{2}[m - 2]^2,$$

which is

$$= [m]^2 - 2 + M, \quad = S - 2.$$

In the remaining cases we may use the theorem to find the number of points in which the generating line meets the Nodal Residue. Using Π as the symbol for the points in question ($\Pi(m, n, p)$ for the scroll $S(m, n, p)$, &c.), we find

61. For the scroll $S(m, n, p)$,

$$(mn - 1) + (np - 1) + (mp - 1) + \Pi(m, n, p) = S - 2 = 2mnp - 2,$$

which gives

$$\Pi(m, n, p) = 2mnp - mn - mp - np + 1.$$

This includes the before-mentioned case

$$\Pi(1, m, n) = (m - 1)(n - 1),$$

and the more particular one

$$\Pi(1, 1, m) = 0.$$

62. For the scroll $S(m^2, n)$,

$$\frac{1}{2}[m]^2 - 1 + M + 2((m - 1)n - 1) + \Pi(m^2, n) = S - 2 = n([m]^2 + M) - 2,$$

which gives

$$\Pi(m^2, n) = n([m]^2 - 2m + 2 + M) - \frac{1}{2}[m]^2 + 1 - M.$$

This includes the before-mentioned particular case

$$\Pi(1, m^2) = \frac{1}{2}[m - 2]^2.$$

63. And lastly for the scroll $S(m^3)$,

$$3\left(\frac{1}{2}[m]^2 - m + 1 + M\right) + \Pi(m^3) = S - 2 = \frac{1}{3}[m]^3 + (m - 2)M - 2,$$

which gives

$$\Pi(m^3) = \frac{1}{3}[m]^3 - \frac{3}{2}[m]^2 + 3m - 5 + M(m - 5).$$

The foregoing expressions for Π might with propriety have been inserted in the Table.

Annex No. 1.—*Investigation of the formula for $S(m^3)$ in the case of the unicursal curve*
(referred to, Art. 39).

Consider the unicursal m -thic curve, the equations whereof are $x : y : z : w = A : B : C : D$, where A, B, C, D are rational and integral functions of a parameter θ ; and let it be required to find the equation of a plane meeting the curve in such manner that three of the points of intersection are in lineá. Taking for the equation of the plane

$$\xi x + \eta y + \zeta z + \omega w = 0,$$

we find between $(\xi, \eta, \zeta, \omega)$ an equation of a certain degree in $(\xi, \eta, \zeta, \omega)$, which is the equation in plane-coordinates of the scroll $S(m^3)$; the degree of the equation is therefore equal to the class of the scroll; but as the class of a scroll is equal to its order, the degree of the equation is equal to the order of the scroll, or say $= S(m^3)$.

Proceeding with the investigation, if θ be determined by the equation

$$\xi A + \eta B + \zeta C + \omega D = 0,$$

then the roots $\theta_1, \theta_2, \dots, \theta_m$ of this equation belong to the points of intersection of the plane and curve; and the corresponding coordinates of these points are $(A_1, B_1, C_1, D_1), (A_2, B_2, C_2, D_2),$ &c.

Suppose that the points 1, 2, 3 are in lineá, and let λ, μ, ν, ρ be the coordinates of an arbitrary point, then the four points are in plano, that is, we have

$$\begin{vmatrix} \lambda & \mu & \nu & \rho \\ A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \end{vmatrix} = 0;$$

and if we form the equation

$$\Pi \begin{vmatrix} \lambda & \mu & \nu & \rho \\ A_1 & B_1 & C_1 & D_1 \\ A_2 & B_2 & C_2 & D_2 \\ A_3 & B_3 & C_3 & D_3 \end{vmatrix} = 0,$$

where Π denotes the product of the terms belonging to all the triads of the m roots, the result will be symmetrical in regard to all the roots; and replacing the symmetrical functions of the roots by their values in terms of the coefficients, we have the required relation between $(\xi, \eta, \zeta, \omega)$.

Π contains $\frac{1}{6}[m]^3$ terms, whereof $\frac{1}{2}[m-1]^2$ contain the m -thic functions (A_1, B_1, C_1, D_1) of the root θ_1 ; that is, the form of Π is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]^3} (\theta_1, 1)^{m \cdot \frac{1}{2}[m-1]^2} (\theta_2, 1)^{m \cdot \frac{1}{2}[m-1]^2} \dots;$$

or, when the symmetrical functions are expressed in terms of the coefficients, the form is

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]^3} (\xi, \eta, \zeta, \omega)^{m \cdot \frac{1}{2}[m-1]^2 \cdot \frac{1}{m}}.$$

Now the above-mentioned determinant is divisible by $(\theta_1 - \theta_2)(\theta_1 - \theta_3)(\theta_2 - \theta_3)$, or Π is divisible by $\prod (\theta_r - \theta_s)(\theta_r - \theta_t)(\theta_s - \theta_t)$; and since this product contains $(3 \times \frac{1}{6}[m]^3 =) \frac{1}{2}[m]^3$ linear factors, and the product $\Delta(\theta_1, \theta_2, \dots, \theta_m)$ of the squared differences of the roots contains $(2 \times \frac{1}{2}[m]^2 =) [m]^2$ linear factors, so that we have

$$\prod (\theta_r - \theta_s)(\theta_r - \theta_t)(\theta_s - \theta_t) = \{\Delta(\theta_1, \theta_2, \dots, \theta_m)\}^{\frac{1}{2}[m-1]^2},$$

and consequently

$$\Delta(\theta_1, \theta_2, \dots, \theta_m) = \text{Disct.} = (\xi, \eta, \zeta, \omega)^{2(m-1)},$$

$$\prod (\theta_r - \theta_s)(\theta_r - \theta_t)(\theta_s - \theta_t) = (\xi, \eta, \zeta, \omega)^{[m-1]^2},$$

so that, omitting this factor, the remaining factor of Π is of the form

$$(\lambda, \mu, \nu, \rho)^{\frac{1}{6}[m]^3} (\xi, \eta, \zeta, \omega)^{\frac{1}{2}[m-1]^2 \cdot \frac{1}{m} \cdot m - [m-1]^2}.$$

But the determinant vanishes if

$$(\lambda, \mu, \nu, \rho) = (A_1, B_1, C_1, D_1), (A_2, B_2, C_2, D_2), (A_3, B_3, C_3, D_3),$$

or say if

$$(\lambda, \mu, \nu, \rho) = (A, B, C, D), \quad \theta = \theta_1, \theta_2, \text{ or } \theta_3;$$

it follows that the product Π contains the factor $(\lambda, \mu, \nu, \rho)$ raised to a corresponding power, or omitting this factor, and observing that

$$\frac{1}{2}[m]^2 - [m-1]^2 - \frac{1}{2}[m]^2 = \frac{1}{2}[m]^2 - [m-1]^2 = \frac{1}{2}[m-1]^2,$$

the remaining factor is of the form

$$(\xi, \eta, \zeta, \omega)^{\frac{1}{3}[m]^3 - (m-1)(m-2)},$$

or we have finally

$$S(m^3) = \frac{1}{3}[m]^3 - (m-1)^2,$$

which is the required expression.

I give the following investigation of the expression $\frac{1}{2}[m-1]^2$ for the number of apparent double points. Imagine through the point $(x=0, y=0, z=0)$ a line cutting the curve in the two points corresponding to the values θ_1, θ_2 of the parameter. We have

$$\frac{A_1}{A_2} = \frac{B_1}{B_2} = \frac{C_1}{C_2},$$

which equations determine θ_1 and θ_2 .

Writing the equations under the form

$$\frac{A_1 B_2 - A_2 B_1}{\theta_1 - \theta_2} = 0, \quad \frac{A_1 C_2 - A_2 C_1}{\theta_1 - \theta_2} = 0,$$

and treating θ_1 and θ_2 as coordinates, each of these equations belongs to a curve of the order $2(m-1)$, having a $(m-1)$ thic point at infinity on each of the axes. The number of intersections thus is

$$= 4(m-1)^2 - (m-1)^2 - (m-1)^2 = 2(m-1)^2.$$

But among these are included points not belonging to the original system, viz. the points for which $(A_1 = 0, A_2 = 0)$ other than those for which $\theta_1 = \theta_2$; the points so included are in number $= m^2 - m$; and omitting them, the number is

$$(2(m-1)^2 - m(m-1)) = [m-1]^2,$$

which is the number of points θ_1 lying in lineá with the origin and another point θ_2 ; the number of apparent double points is the half of this, or $h = \frac{1}{2}[m-1]^2$. And thence

$$M = (-\frac{1}{2}[m]^2 + h) - (m-1).$$

I investigate also the number of lines through two points which meet two arbitrary lines; this is in fact $= S(1, m^2)$, which for the curve in question is

$$= ([m]^2 - (m-1)^2) = (m-1)^2.$$

Let the equations of the two lines be $(x = 0, y = 0)$ and $(z = 0, w = 0)$; then the conditions to be satisfied are

$$\frac{A_1}{A_2} = \frac{B_1}{B_2}, \quad \frac{C_1}{C_2} = \frac{D_1}{D_2};$$

or writing these under the form

$$\frac{A_1 B_2 - A_2 B_1}{\theta_1 - \theta_2} = 0, \quad \frac{C_1 D_2 - C_2 D_1}{\theta_1 - \theta_2} = 0,$$

and treating θ_1, θ_2 as coordinates, the number of intersections of these two curves is $= 2(m-1)^2$, the same as for the two curves last above considered. And the number of the lines in question is one half of this, or $= (m-1)^2$.

Lemma employed in the following Annexes 2 and 3. Formulae for the order and weight of certain systems of equations.

Let $a_{\alpha, \alpha'}$ denote a function of the degree α in the order variables $(x, y, \dots)$, and of the degree α' in the weight variables $(x', y', \dots)$, and so in other cases; and consider first the equation

$$\begin{vmatrix} a_{\alpha, \alpha'} & (\alpha + A)_{\alpha+A, \alpha'} & \cdots \\ \beta_{\beta, \beta'} & (\beta + A)_{\beta+A, \beta'} & \cdots \\ \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square; then

$$\text{Order} = \Sigma\alpha + \Sigma A,$$

$$\text{Weight} = \Sigma\alpha' + \Sigma A'.$$

Consider next the system

$$\begin{vmatrix} a_{\alpha, \alpha'} & (\alpha + A)_{\alpha+A, \alpha'} & (\alpha + B)_{\alpha+B, \alpha'} & \cdots \\ \beta_{\beta, \beta'} & (\beta + A)_{\beta+A, \beta'} & (\beta + B)_{\beta+B, \beta'} & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square +1, that is, the number of columns exceeds by 1 the number of lines; then

$$\text{Order} = \Sigma AB - \Sigma\alpha\beta + \Sigma\alpha(\Sigma A + \Sigma\alpha),$$

$$\text{Weight} = (\Sigma A + \Sigma \alpha)(\Sigma A' + \Sigma \alpha') - \Sigma A A' + \Sigma \alpha \alpha'.$$

And again, the system

$$\begin{vmatrix} a_{\alpha, \alpha'} & (\alpha + A)_{\alpha+A, \alpha'} & (\alpha + B)_{\alpha+B, \alpha'} & (\alpha + C)_{\alpha+C, \alpha'} & \cdots \\ \beta_{\beta, \beta'} & (\beta + A)_{\beta+A, \beta'} & (\beta + B)_{\beta+B, \beta'} & (\beta + C)_{\beta+C, \beta'} & \cdots \\ \vdots & \vdots & \vdots & \vdots & \ddots \end{vmatrix} = 0,$$

where the matrix is a square +2, that is, the number of columns exceeds by 2 the number of lines; then

$$\text{Order} = \Sigma ABC + \Sigma \alpha \beta \gamma + \Sigma \alpha (\Sigma AB - \Sigma \alpha \beta) + ((\Sigma \alpha)^2 - \Sigma \alpha \beta)(\Sigma A + \Sigma \alpha),$$

$$\text{Weight} = \{ \Sigma AB - \Sigma \alpha \beta + \Sigma \alpha (\Sigma A + \Sigma \alpha) \} (\Sigma A' + \Sigma \alpha') - (\Sigma A + \Sigma \alpha)(\Sigma A A' - \Sigma \alpha \alpha') + \Sigma A^2 A' + \Sigma \alpha^2 \alpha'.$$

The last formula, for the weight of the square +2 system, was communicated to me by Dr Salmon; the others are all in effect given in the Appendix, “On the Order of Systems of Equations,” to his Treatise on the Analytic Geometry of Three Dimensions; and in the investigation in the following Annexes 2 and 3, the route which I have followed was completely traced out for me by him, so that I have only supplied the details of the work.

Annex No. 2.—*Investigation of the formula for $S(m^3)$, when the curve m is the pq complete intersection, viz. when it is the intersection of two surfaces of the orders p and q respectively (referred to, Art. 40).*

Let $U = 0, V = 0$ be the equations of the two surfaces of the orders p and q respectively. Take (x, y, z, w) the coordinates of a point on the curve, so that for these coordinates we have $U = 0, V = 0$; and in the equations of the two curves respectively, write for the coordinates $x + \rho x', y + \rho y', z + \rho z', w + \rho w'$; then putting for shortness

$$\Delta = x' \partial_x + y' \partial_y + z' \partial_z + w' \partial_w,$$

the resulting equations may be represented by

$$(\Delta U, \Delta^2 U, \dots, \Delta^p U \mid 1, \rho)^{p-1} = 0,$$

$$(\Delta V, \Delta^2 V, \dots, \Delta^q V \mid 1, \rho)^{q-1} = 0,$$

where it is to be noticed that besides the expressed literal coefficients there are numerical coefficients (not as the notation usually denotes, the binomial coefficients, but) $= \frac{1}{1}, \frac{1}{1 \cdot 2}, \frac{1}{1 \cdot 2 \cdot 3}, \&c.$

Supposing that (x', y', z', w') are the current coordinates of a point on the line drawn through the point (x, y, z, w) to meet the curve in two other points, the equations in ρ must have two common roots, and this gives a system equivalent to two equations, or say a plexus of two equations. If from the plexus and the two equations $U = 0, V = 0$ we eliminate (x, y, z, w) , we obtain an equation $S' = 0$ in (x', y', z', w') , which is in fact the equation of the scroll $S(m^3)$, taken (as is easily seen to be the case) thrice; that is, $S(m^3) = \frac{1}{3} \text{Deg. } S'$. But observing that the coordinates (x', y', z', w') enter into the plexus only and not into the functions U, V , and treating (x', y', z', w') as weight variables, $\text{Deg. } S' = \text{Weight of System } (U = 0, V = 0, \text{ Plexus}) = \text{Deg. } U \times \text{Deg. } V \times \text{Weight of Plexus} = pq \times \text{Weight of Plexus}$; or, writing $pq = \beta$,

$$S(m^3) = \frac{1}{3} \beta \times \text{Weight of Plexus}.$$

The plexus in question is the square +1 system

$$\begin{vmatrix} \Delta U, \Delta^2 U, \dots \\ \Delta V, \Delta^2 V, \dots \end{vmatrix} = 0,$$

$p + q - 3$ columns, $(q - 2) + (p - 2) = (p + q - 4)$ lines; or representing the terms according to their order and weight, that is, degree in (x, y, z, w) and (x', y', z', w') respectively (the order and weight of the evanescent terms being fixed so as that they may form a regular series with the other terms), the system is, with $p + q - 3$ columns:

$$\begin{aligned} (p - 2) \text{ lines: } & (p - 1)_1, (p - 2)_2, \dots, (p - 1)_{p-1}, \dots, 0, \\ (q - 2) \text{ lines: } & (q - 1)_1, (q - 2)_2, \dots, (q - 1)_{q-1}, \dots, 0, \end{aligned}$$

so that

$$\begin{aligned} \alpha, \beta, \dots &= p - 1, p - 1, \dots, p + q - 5, q - 1, q, \dots, p + q - 5, \\ \alpha', \beta', \dots &= 1, 0, \dots, -(p + q - 4), 1, 0, \dots, -(p + q - 4), \\ A, B, \dots &= -1, -2, \dots, -(p + q - 4), \\ A', B', \dots &= 1, 2, \dots, p + q - 4, \end{aligned}$$

or, as regards the first two lines,

$$\alpha, \beta = p - 1, q - 1, \quad \alpha', \beta' = 1, 1.$$

We then find

$$\begin{aligned} \Sigma\alpha &= 2(q - 2) + \frac{1}{2}(q - 2)(q - 1) + 2(p - 2) + \frac{1}{2}(p - 2)(p - 1), \\ \Sigma\alpha' &= \frac{1}{2}(q - 2)(q - 1) + (p - 2)(q - 2) + \frac{1}{2}(p - 2)(p - 1), \\ \Sigma A &= -\Sigma A' = -\frac{1}{2}(p + q - 4)(p + q - 3), \\ \Sigma\alpha\alpha' &= \frac{1}{2}(p - 2)(q - 2) - (p - 4) \cdot \frac{1}{2}(q - 2)(q - 1) - \frac{1}{2}(q - 2)(q - 1)(2q - 3) \\ &\quad + \frac{1}{2}(q - 2)(p - 2) - (q - 4) \cdot \frac{1}{2}(p - 2)(p - 1) - \frac{1}{2}(p - 2)(p - 1)(2p - 3), \\ \Sigma AA' &= -\frac{1}{6}(p + q - 4)(p + q - 3)(2p + 2q - 7). \end{aligned}$$

Which, putting therein $p + q = \alpha$, $pq = \beta$, give

$$\begin{aligned} \Sigma\alpha &= \beta + \frac{1}{2}\alpha^2 - \frac{11}{2}\alpha + 10, \\ \Sigma\alpha' &= \beta - \frac{1}{2}\alpha^2 + \frac{5}{2}\alpha - 10, \\ \Sigma A &= -\Sigma A' = -\frac{1}{2}\alpha^2 + \frac{7}{2}\alpha - 6, \\ \Sigma\alpha\alpha' &= \frac{1}{2}\alpha\beta - \frac{1}{6}\alpha^3 + \frac{3}{2}\alpha^2 - \frac{31}{3}\alpha + 26, \\ \Sigma AA' &= -\frac{1}{6}\alpha^3 + \frac{3}{2}\alpha^2 - \frac{52}{6}\alpha + 14, \end{aligned}$$

and therefore

$$\begin{aligned} \Sigma A + \Sigma\alpha &= \beta - 2\alpha + 4, \\ \Sigma A' + \Sigma\alpha' &= \beta - 4, \end{aligned}$$

and consequently

$$\Sigma AA' - \Sigma\alpha\alpha' = -\frac{1}{2}\alpha\beta + 5\alpha - 12,$$

which is right.

$$\begin{aligned} \text{Weight} &= (\beta - 2\alpha + 4)(\beta - 4) + \frac{1}{2}\alpha\beta - 5\alpha + 12 \\ &= \beta^2 - \frac{3}{2}\alpha\beta + 6\alpha - 8 \\ &= \frac{1}{2}(\beta - 2)(2\beta - 3\alpha + 4), \end{aligned}$$

and

$$\begin{aligned} S(m^3) &= \frac{1}{3}\beta \times \text{Weight} \\ &= \frac{1}{6}\beta(\beta - 2)(2\beta - 3\alpha + 4). \end{aligned}$$

Annex No. 3.—*Investigation of $G(m^4)$ in the case where the curve m is a pq complete intersection* (referred to, Art. 42).

Suppose, as before, that $U = 0, V = 0$ are the equations of the two surfaces of the orders p and q respectively; taking also (x, y, z, w) as the coordinates of a point on the curve, and substituting in the equations $x + \rho x', y + \rho y', z + \rho z', w + \rho w'$ in place of the coordinates, then if $\Delta = x' \partial_x + y' \partial_y + z' \partial_z + w' \partial_w$, we have as before

$$(\Delta U, \Delta^2 U, \dots, \Delta^p U \mid 1, \rho)^{p-1} = 0,$$

$$(\Delta V, \Delta^2 V, \dots, \Delta^q V \mid 1, \rho)^{q-1} = 0,$$

where the numerical coefficients $\frac{1}{1}, \frac{1}{1 \cdot 2}, \&c.$ are to be understood as before.

Suppose now that (x, y, z, w) are the coordinates of a point on the curve, through which point there passes a line through three other points, or line $G(m^4)$; and that (x', y', z', w') are the current coordinates of a point on such line; the two equations in ρ must have three equal roots; or we must have a system equivalent to three equations, or say a plexus of three equations. The coordinates (x', y', z', w') , although four in number, are in fact eliminable from this plexus; or what is the same thing, combining with the plexus the equation

$$\alpha x' + \beta y' + \gamma z' + \delta w' = 0$$

of an arbitrary plane, and then eliminating (x', y', z', w') , the result is of the form

$$(\alpha x + \beta y + \gamma z + \delta w)^\theta D = 0,$$

where D is a function of (x, y, z, w) only; and considering (x, y, z, w) as weight variables, $\theta = \text{Order of Plexus}$. But degree in (x, y, z, w) of $(\alpha x + \beta y + \gamma z + \delta w)^\theta D$ is = Weight of Plexus, and therefore Degree of D is = Weight of Plexus $-\theta$, = (Weight $-\text{Order}$) of Plexus.

The equations $U = 0, V = 0, D = 0$ then give the coordinates (x, y, z, w) of the points through which may be drawn a line $G(m^4)$; viz. they give (as it is easy to see) these points four times over. And we therefore have

$$\begin{aligned} G(m^4) &= \frac{1}{4} \text{Order } (U = 0, V = 0, D = 0) \\ &= \frac{1}{4} \text{Deg. } U \cdot \text{Deg. } V \cdot \text{Deg. } D \\ &= \frac{1}{4} \beta \times (\text{Weight} - \text{Order}) \text{ of Plexus.} \end{aligned}$$

The Plexus is here the square +2 system

$$\left| \begin{array}{c} \Delta U, \Delta^2 U, \dots \\ \Delta V, \Delta^2 V, \dots \end{array} \right| = 0,$$

$p + q - 4$ columns, $(q - 3) + (p - 3) = p + q - 6$ lines, the leading case being $\theta = 0$; representing the terms by their order and weight (the weight variables being (x, y, z, w) , and the order variables (x', y', z', w')), and attributing as before an order and weight to the evanescent terms, the system is, with $p + q - 4$ columns:

$$\begin{array}{c} 1, 2, \dots, p - 1, \\ 1, 2, \dots, q - 1, \end{array}$$

so that we have

$$\begin{aligned} \alpha, \beta, \dots &= 1, 0, -1, \dots, -(q - 5), \quad 1, 0, -1, \dots, -(p - 5), \\ \alpha', \beta', \dots &= p - 1, p, p + 1, \dots, p + q - 5, \quad q - 1, q, q + 1, \dots, q + p - 5, \\ A, B, \dots &= 1, 2, \dots, p + q - 5, \\ A', B', \dots &= -1, -2, \dots, -(p + q - 5), \end{aligned}$$

or, as regards the first two lines,

$$\alpha, \beta, \dots = 2 - e, 2 - \phi, \quad e = 1 \text{ to } q - 3, \quad \phi = 1 \text{ to } p - 3,$$

$$\alpha', \beta', \dots = p - 2 + e, p - 2 + \phi.$$

We then find

$$\begin{aligned} \Sigma\alpha &= 2(q-3) - \frac{1}{2}(q-3)(q-2) + 2(p-3) - \frac{1}{2}(p-3)(p-2), \\ \Sigma\alpha' &= (p-2)(q-3) + \frac{1}{2}(q-3)(q-2) + (q-2)(p-3) + \frac{1}{2}(p-3)(p-2), \\ \Sigma\alpha^2 &= 4(q-3) - 4 \cdot \frac{1}{2}(q-3)(q-2) + \frac{1}{2}(q-3)(q-2)(2q-5) \\ &\quad + 4(p-3) - 4 \cdot \frac{1}{2}(p-3)(p-2) + \frac{1}{2}(p-3)(p-2)(2p-5), \\ \Sigma\alpha^3 &= 8(q-3) - 12 \cdot \frac{1}{2}(q-3)(q-2) + 6 \cdot \frac{1}{2}(q-3)(q-2)(2q-5) - \frac{1}{4}(q-3)^2(q-2)^2 \\ &\quad + 8(p-3) - 12 \cdot \frac{1}{2}(p-3)(p-2) + 6 \cdot \frac{1}{2}(p-3)(p-2)(2p-5) - \frac{1}{4}(p-3)^2(p-2)^2, \\ \Sigma\alpha\alpha' &= 2(p-2)(q-3) - (p-4) \cdot \frac{1}{2}(q-3)(q-2) - \frac{1}{6}(q-3)(q-2)(2q-5) \\ &\quad + 2(q-2)(p-3) - (q-4) \cdot \frac{1}{2}(p-3)(p-2) - \frac{1}{6}(p-3)(p-2)(2p-5), \\ \Sigma\alpha^2\alpha' &= 4(p-2)(q-3) - 4(p-3) \cdot \frac{1}{2}(q-3)(q-2) + (p-6) \cdot \frac{1}{2}(q-3)(q-2)(2q-5) + \frac{1}{4}(q-3)^2(q-2)^2 \\ &\quad + 4(q-2)(p-3) - 4(q-3) \cdot \frac{1}{2}(p-3)(p-2) + (q-6) \cdot \frac{1}{2}(p-3)(p-2)(2p-5) + \frac{1}{4}(p-3)^2(p-2)^2, \\ \Sigma A &= \frac{1}{2}(p+q-5)(p+q-4), \\ \Sigma A^2 &= -\Sigma AA' = \frac{1}{6}(p+q-5)(p+q-4)(2p+2q-9), \\ \Sigma A^3 &= -\Sigma A^2 A' = \frac{1}{4}(p+q-5)^2(p+q-4)^2, \end{aligned}$$

which, putting therein $p+q = \alpha$, $pq = \beta$, and from the reduced expressions obtaining the values of $\Sigma\alpha\beta$, &c., give

$$\begin{aligned} \Sigma\alpha &= \beta - \frac{1}{2}\alpha^2 + \frac{9}{2}\alpha - 18, \\ \Sigma\alpha^2 &= \beta(-\alpha+9) + \frac{1}{3}\alpha^3 - \frac{17}{2}\alpha^2 + \frac{161}{3}\alpha - 88, \\ \Sigma\alpha^3 &= \beta^2(-\frac{3}{2}) + \beta(\alpha^2 - \frac{31}{2}\alpha + \frac{161}{2}) - \frac{1}{4}\alpha^4 + \frac{27}{2}\alpha^3 - \frac{691}{4}\alpha^2 + 939\alpha - 198, \\ \Sigma\alpha\beta &= \beta^2(\frac{1}{2}) + \beta(-\frac{3}{2}\alpha^2 + 5\alpha - \frac{37}{2}) + \frac{1}{6}\alpha^3 - \frac{17}{6}\alpha^2 - \frac{169}{6}\alpha^2 - \frac{169}{6}\alpha + 191, \\ \Sigma\alpha^2\alpha' &= \beta^2(\alpha) + \beta(-\frac{1}{2}\alpha^2 + 9\alpha - \frac{159}{2}) + \beta\alpha(\alpha^2 - \frac{9}{2}\alpha + \frac{29}{2}\alpha - \frac{20}{2}) \\ &\quad - \frac{1}{2}\alpha^5 + \frac{25}{4}\alpha^4 - \frac{125}{4}\alpha^3 + \frac{205}{2}\alpha^2 - \frac{289}{2}\alpha^2 + 1071\alpha - 1560, \\ \Sigma\alpha' &= \beta + \frac{1}{2}\alpha^2 - \frac{11}{2}\alpha + 18, \\ \Sigma\alpha\alpha' &= \beta(-\frac{1}{2}\alpha) - \frac{1}{3}\alpha^3 + \frac{7}{2}\alpha^2 - \frac{47}{3}\alpha + 58, \\ \Sigma\alpha^2\alpha' &= \beta^2(-\frac{1}{2}) + \beta(-\frac{1}{2}\alpha^2 + 9\alpha - \frac{151}{2}) + \frac{1}{4}\alpha^4 - \frac{17}{2}\alpha^3 + \frac{161}{2}\alpha^2 - 119\alpha + 198, \\ \Sigma A &= \frac{1}{2}\alpha^2 - \frac{9}{2}\alpha + 10, \\ \Sigma A^2 &= -\Sigma AA' = \frac{1}{3}\alpha^3 - \frac{9}{2}\alpha^2 + \frac{55}{3}\alpha - 30, \\ \Sigma A^3 &= -\Sigma A^2 A' = \frac{1}{4}\alpha^4 - \frac{17}{2}\alpha^3 + \frac{161}{4}\alpha^2 - 90\alpha + 100, \\ \Sigma AB &= \frac{1}{6}\alpha^3 - \frac{11}{4}\alpha^2 + \frac{43}{2}\alpha^2 - \frac{27}{2}\alpha + 65, \\ \Sigma ABC &= \frac{1}{12}\alpha^5 - \frac{11}{6}\alpha^4 + \frac{59}{4}\alpha^3 - \frac{121}{2}\alpha\alpha^2 + \frac{139}{2}\alpha^2 - \frac{1303}{2}\alpha + \frac{1267}{2}, \end{aligned}$$

we then find

$$\begin{aligned}
\Sigma A + \Sigma \alpha &= \beta - 8, \\
\Sigma A' + \Sigma \alpha' &= \beta - 3\alpha + 8, \\
\Sigma AB - \Sigma \alpha \beta &= \beta^2 \left(-\frac{1}{2}\right) + \beta \left(\frac{3}{2}\alpha^2 - 5\alpha + \frac{37}{2}\right) - 4\alpha^2 + 36\alpha - 126, \\
\Sigma AA' - \Sigma \alpha \alpha' &= \beta \left(-\frac{1}{2}\alpha\right) + 9\alpha - 28, \\
&\quad - \alpha^4 + \frac{17}{2}\alpha^3 - 162\alpha^2 + \frac{169}{6}\alpha - 1210, \\
\Sigma A^2 A' + \Sigma \alpha^2 \alpha' &= \beta^2 \left(-\frac{1}{2}\right) + \beta \left(-\frac{1}{2}\alpha^2 + 9\alpha - \frac{151}{2}\right) - 29\alpha + 98;
\end{aligned}$$

and then also

$$\begin{aligned}
\Sigma \alpha (\Sigma A + \Sigma \alpha) &= \beta^2 + \beta \left(-\frac{1}{2}\alpha^2 + \frac{5}{2}\alpha - 26\right) + 4\alpha^2 - 36\alpha + 144, \\
(\Sigma AB - \Sigma \alpha \beta) + \Sigma \alpha (\Sigma A + \Sigma \alpha) &= \beta^2 \left(\frac{1}{2}\right) + \beta \left(\alpha^2 - \frac{5}{2}\alpha - \frac{15}{2}\right) + 18, \\
\{(\Sigma AB - \Sigma \alpha \beta) + \Sigma \alpha (\Sigma A + \Sigma \alpha)\}(\Sigma A' + \Sigma \alpha') &= \beta^3 \left(\frac{1}{2}\right) + \beta^2 \left(-\alpha + \frac{5}{2}\right) + \beta \left(\alpha^2 - 13\alpha + 28\right) + 72\alpha - 224, \\
-(\Sigma A + \Sigma \alpha)(\Sigma AA' - \Sigma \alpha \alpha') &= \beta \left(\frac{1}{2}\alpha\right) + \beta \left(-\frac{15}{2}\alpha + 28\right) + \dots,
\end{aligned}$$

and

$$\Sigma A^2 A' + \Sigma \alpha^2 \alpha' = (\text{ut supra}) \beta^2 \left(-\frac{1}{2}\right) + \beta \left(-\frac{1}{2}\alpha^2 + 9\alpha - \frac{151}{2}\right) - 29\alpha + 98;$$

whence, adding the last three expressions, we find

$$\text{Weight} = \beta^3 \left(\frac{1}{2}\right) + \beta^2 \left(-\frac{3}{2}\alpha + \frac{1}{2}\right) + \beta \left(\frac{3}{2}\alpha^2 + \frac{3}{2}\alpha - \frac{15}{2}\right) - 11\alpha + 18;$$

and for the order we have

$$(\Sigma \alpha)^2 - \Sigma \alpha \beta = \beta^2 \left(\frac{1}{2}\right) + \beta \left(-\frac{1}{2}\alpha^2 + 4\alpha - \frac{55}{2}\right) + \frac{1}{4}\alpha^4 - \frac{17}{2}\alpha^3 + \frac{161}{2}\alpha^2 - \frac{161}{2}\alpha + 133;$$

and then

$$\begin{aligned}
\Sigma ABC + \Sigma \alpha \beta \gamma &= (\text{ut supra}), \\
(\Sigma AB - \Sigma \alpha \beta) \Sigma \alpha &= -\alpha^4 + \frac{17}{2}\alpha^3 - 162\alpha^2 + \frac{169}{6}\alpha - 1210, \\
\beta \left(-\frac{1}{2}\right) + \beta^2 \left(\frac{1}{2}\alpha^2 - \frac{31}{2}\alpha + \frac{161}{2}\right) &+ \beta \left(-\frac{1}{4}\alpha^4 + \frac{169}{4}\alpha^3 - \frac{91}{2}\alpha^2 + \frac{205}{4}\alpha - 531\right) \\
&+ 2\alpha^5 - 36\alpha^4 + 297\alpha^3 - 1215\alpha + 2268, \\
((\Sigma \alpha)^2 - \Sigma \alpha \beta)(\Sigma A + \Sigma \alpha) &= \beta^3 \left(\frac{1}{2}\right) + \beta^2 \left(-\frac{1}{2}\alpha^2 + 4\alpha - \frac{55}{2}\right) + \beta \left(-\alpha^4 + \frac{17}{2}\alpha^3 - 135\alpha^2 + \frac{169}{2}\alpha - 106\right)
\end{aligned}$$

whence, adding these three expressions,

$$\text{Order} = \beta^3 \left(\frac{1}{2}\right) + \beta^2 \left(-\frac{3}{2}\alpha + \frac{1}{2}\right) + \beta \left(\frac{3}{2}\alpha^2 - \frac{3}{2}\alpha + \frac{15}{2}\right) - 6;$$

and by means of the foregoing expression for the weight, we then have

$$\text{Weight} - \text{Order} = \beta^2 \left(\frac{1}{2}\right) + \beta \left(-\alpha\right) + \beta \left(\frac{1}{2}\alpha^2 + 3\alpha - \frac{15}{2}\right) - 11\alpha + 24;$$

and therefore

$$\begin{aligned}
G(m^4) &= \frac{1}{4}\beta \times (\text{Weight} - \text{Order}), \\
&= \frac{1}{12}\beta \left(2\beta^3 + \beta^2 \left(-6\alpha\right) + \beta \left(3\alpha^2 + 18\alpha - 26\right) - 66\alpha + 144\right),
\end{aligned}$$

which is right.

Annex No. 4.—*Order of Torse*(m, n) (referred to, Art. 44).

We have to find the order of the developable or Torse generated by a line meeting two curves of the orders m, n respectively; viz. representing by μ, ν the classes of the two curves respectively, it is to be shown that the expression for the Order is

$$\text{Torse}(m, n) = m\nu + n\mu.$$

I remark, in the first place, that, given two surfaces of the orders p and q respectively, the curve of intersection is of the order pq and class $pq(p+q-2)$, or as this may be written, class $= qp(p-1) + pq(q-1)$. Reciprocally for two surfaces of the classes p and q respectively, the Torse enveloped by their common tangent planes is of the class pq and order $qp(p-1) + pq(q-1)$. Now, in the same way that a surface of the order p may degenerate into a Torse of the order p , so a surface of the class p may degenerate into a curve of the class p ; and the class of a curve being p , then (disregarding singularities) its order is $= p(p-1)$. Hence replacing p and $p(p-1)$ by μ and m respectively, and in like manner q and $q(q-1)$ by ν and n respectively, we have $m\nu + n\mu$ as the order of the Torse generated by the tangent planes of the curves of the orders m and n respectively; where by tangent plane of a curve is to be understood a plane passing through a tangent line of the curve. The intersection of two consecutive tangent planes is a line meeting the two curves, which line is the generating line of the Torse, and such Torse is therefore the Torse (m, n) in question.

The foregoing investigation is not very satisfactory, but I confirm it by considering the case of two plane curves, orders m and n , and classes μ and ν , respectively. The tangents of the two curves can, it is clear, only meet on the line of intersection of the planes of the curves; and the construction of the Torse is in fact as follows: from any point of the line of intersection draw a tangent to m and a tangent to n , then the line joining the points of contact of these tangents is a generating line of the Torse. The order of the Torse is equal to the number of generating lines which meet an arbitrary line; and taking for the arbitrary line the line of intersection of the two planes, it is easy to see that the only generating lines which meet the line of intersection are those for which one of the points of contact lies on the line of intersection; that is, they are the generating lines derived from the points in which the line of intersection meets one or other of the two curves; they are therefore in fact the tangents drawn to the curve n from the points in which the line of intersection meets the curve m , together with the tangents drawn to the curve m from the points in which the line of intersection meets the curve n . Now the line meets the curve n in n points, and from each of these there are μ tangents to the curve m ; and it meets the curve m in m points, and from each of these there are ν tangents to the curve n ; hence the entire number of the tangents in question is $= n\mu + m\nu$, which confirms the theorem.

Annex No. 5.—*Order of Torse*(m^2) (referred to, Art. 46).

We have here to find the order of the developable or Torse generated by a line meeting a curve of the order m twice, viz. the class of the curve being μ , it is to be shown that we have

$$\text{Torse}(m^2) = (m-3)\mu.$$

I deduce the expression from the formula given p. 424 of Dr Salmon's 'Geometry of Three Dimensions;' viz. putting in his formula $\beta = 0$, and μ for his r , we have

$$\text{Order} = m(\mu-4) - \frac{1}{2}\alpha = m\mu - (4m + \frac{1}{2}\alpha),$$

where (see p. 234 et seq.)

$$\frac{1}{2}\alpha = (n-m) = 3m(m-2) - 6h - m,$$

and thence

$$\mu = m(m-1) - 2h,$$

so that we have

$$3\mu - \frac{1}{2}\alpha = 4m, \quad \text{or} \quad 4m + \frac{1}{2}\alpha = 3\mu,$$

$$\text{Order} = (m-3)\mu.$$

A more complete discussion of the Torses (m, n) and (m^2) is obviously desirable; but as they are only incidentally connected with the subject of the present memoir, I have contented myself with obtaining the required results in the way which most readily presented itself.

340. A Second Memoir on Skew Surfaces, otherwise Scrolls

[From the Philosophical Transactions of the Royal Society of London, vol. CLIV. (for the year 1864), pp. 559–576. Received April 29,—Read May 26, 1864.]

The principal object of the present memoir is to establish the different kinds of skew surfaces of the fourth order, or Quartic Scrolls; but, as preliminary thereto, there are some general researches connected with those in my former memoir “On Skew Surfaces, otherwise Scrolls”¹; and I also reproduce the theory (which may be considered as a known one) of cubic scrolls; there are also some concluding remarks which relate to the general theory. As regards quartic scrolls, I remark that M. Chasles, in a footnote to his paper, “Description des courbes de tous les ordres situées sur les surfaces réglées du troisième et du quatrième ordres”², states, “les surfaces réglées du quatrième ordre... admettent quatorze espèces.” This does not agree with my results, since I find only eight species of quartic scrolls; the developable surface or “torse” is perhaps included as a “surface réglée;” but as there is only one species of quartic torse, the deficiency is not to be thus accounted for. My enumeration appears to me complete, but it is possible that there are subforms which M. Chasles has reckoned as distinct species.

On the Degeneracy of a Scroll, Article Nos. 1 to 5.

1. A scroll considered as arising from any geometrical construction, for instance one of the scrolls $S(m, n, p)$, $S(m^2, n)$, $S(m^3)$ considered in my former memoir, or say in general the scroll S , may break up into two or more inferior scrolls S' , S'' , ...; but as long as S' , S'' , ... are proper scrolls (not torses, and *a fortiori* not cones or planes), no one of these can be considered, apart from the others, as the result of the geometrical

[Paper 340 continues on p. 202; this excerpt ends at p. 201.]

[End of excerpt: pp. 177–201 of Vol. V; PDF pp. 201–225.]

¹Philosophical Transactions, vol. CLIII. (1863), pp. 453–483, [339].

²Comptes Rendus, t. LIII. (1861), see p. 888.